XXXXX University

Doctoral Program in XXXXX

Provisional Research Title

Runtime Validity Estimation for the Integration of Hybrid Learned Models into Vehicle Dynamics Control

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Year

Abstract

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Chapter 1

Introduction

To illustrate the problem, consider a driver traveling along a highway at 100 km/h when it starts to rain. As visibility decreases, the driver decides to pull over onto the shoulder. During the maneuver, however, water on the road may reduce tire–road grip, making it harder to change course. What began as an attempt to stop the journey can, under these conditions, lead to a loss of control and a risk of collision with other vehicles or roadside barriers.

This example highlights a difficulty that extends beyond the driver's decision: the ability to execute a maneuver depends on the contact conditions between the tires and the road. When these conditions change, the forces available for braking or changing direction may also change. Understanding how grip varies along the route is therefore part of the problem of keeping the vehicle on its intended path.

Pavement wear and water on the road are among the factors that affect grip. Over time, traffic polishes the aggregates and changes the surface texture, reducing skid resistance [8, 4, 9]. During rainfall, water can impair tire–road contact and, under hydroplaning conditions, cause that contact to be lost [6]. Ruts in the wheel paths allow water to accumulate, contributing to the problem [5]. Both pavement wear and the conditions encountered during a journey can therefore alter the grip available for a maneuver.

This limitation becomes relevant when a driver needs to change lanes at short notice or swerve to avoid an obstacle. In these maneuvers, the lateral forces required to follow the intended path may exceed those that the tires can transmit to the road [1]. The combination of high speed and low grip can then cause lateral sliding and yaw instability, leading to a loss of vehicle control [3]. The challenge is therefore to execute the maneuver within the limits imposed by tire–road interaction.

Vehicle control systems address this challenge by assisting the driver with braking and maintaining vehicle stability. The anti-lock braking system (ABS) and electronic stability control (ESC) illustrate this development: the former aims to prevent wheel lock during braking, while the latter helps correct deviations between vehicle motion and the driver's intended direction. The introduction of ESP, an implementation of electronic stability

control launched by Bosch in 1995, extended this assistance to situations involving a loss of stability [2]. Development continues through the integration of braking, steering, and propulsion, whose coordinated operation draws on the capabilities of each system to regulate vehicle motion [7]. For this assistance to respond to the situations encountered while driving, it remains necessary to account for how changes in grip affect the vehicle's ability to follow a path [3].

1.1 Context and motivation

1.2 Research problem

1.3 Scope of the problem (Priority 1)

1.4 Research gap

1.5 Research question

How can the runtime validity of a learned vehicle dynamics residual model be estimated and used to adapt an MPC/NMPC controller under operating conditions not represented during training?

1.6 Specific research questions

RQ1: Which types of physically relevant shifts in vehicle dynamics cause a significant loss of validity in the learned residual?

RQ2: What information do epistemic uncertainty, aleatoric uncertainty, calibration, and an OOD indicator provide about control-relevant model error?

RQ3: To what extent are uncertainty, calibration, and OOD signals complementary or redundant in characterizing runtime validity?

RQ4: How can a confidence/validity signal $C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Calibration}, \text{OOD})$ be constructed to reflect the error relevant to the controller?

RQ5: How should C_k be used to adapt the MPC/NMPC horizon to reduce closed-loop performance degradation when the validity of the learned model decreases?

RQ6: What improvement does the proposed architecture offer over a controller based only on the physical model, a controller that uses the learned model without validity monitoring, and architectures that use only validity or only OOD information?

RQ7: As an optional extension, can validity information be used to modify the action of a CBF or trigger data/model update strategies in response to epistemic uncertainty?

1.7 Research hypothesis

A calibrated combination of epistemic uncertainty, aleatoric uncertainty, and evidence of out-of-distribution operation can provide a useful estimate of the runtime validity of a learned vehicle dynamics residual. Incorporating this estimate into the adaptation of an MPC/NMPC controller can reduce closed-loop performance degradation under regime changes compared with architectures that use the learned model without validity awareness.

Chapter 2

Objectives

2.1 Overall objective

Develop and evaluate a methodology for real-time validity estimation for a hybrid vehicle dynamics model comprising a physical model and a learned residual, and integrate this estimate into an MPC/NMPC controller to reduce closed-loop performance degradation under operating conditions not represented during training.

2.2 Specific objectives

SO1: Formulate a hybrid vehicle dynamics architecture comprising a control-oriented physical model and a learned residual.

SO2: Provide operational definitions of the types of shift/OOD relevant to the study, distinguishing at least unrepresented physical regimes and dynamics OOD, and including input OOD if time permits.

SO3: Estimate and calibrate the aleatoric and epistemic uncertainty of the learned model using a heteroscedastic model and a small ensemble, complemented by a computationally inexpensive OOD indicator.

SO4: Experimentally characterize the relationships among distribution shift, uncertainty, calibration, OOD, prediction error, and control-relevant error.

SO5: Define a representation of runtime confidence/validity

$$C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Calibration}, \text{OOD})$$

and evaluate its ability to anticipate or represent model degradation in a way that is useful to the controller.

SO6: Integrate the validity signal into an MPC/NMPC controller through horizon adaptation and evaluate its effect on closed-loop performance and empirical robustness.

SO7: Compare the proposed approach with baseline architectures that isolate the effects of the learned model, validity estimation, and OOD detection.

SO8: As an extension, if the schedule permits, evaluate the use of the validity signal in a CBF and/or a DAgger-based data acquisition and retraining strategy for conditions associated with epistemic uncertainty.

Chapter 3

Fundamentals of vehicle dynamics and control

3.1 Vehicle dynamics relevant to control

3.2 Control-oriented models and model mismatch

3.3 Hybrid models: physics and learned residuals

3.4 Model predictive control

3.5 Control barrier functions as an optional extension

Chapter 4

Learned information and runtime validity

4.1 Learned information relevant to the controller

4.2 Predictive uncertainty

4.2.1 Aleatoric uncertainty

4.2.2 Epistemic uncertainty

4.2.3 Calibration

4.3 OOD and distribution shift in vehicle dynamics

4.3.1 Unrepresented physical regime

4.3.2 Dynamics OOD

4.3.3 Input OOD

4.3.4 Sensor failure or degradation

4.4 Runtime validity and confidence

$$C_k = g(U_{\mathrm{epi},k}, U_{\mathrm{ale},k}, \mathrm{Cal}_k, S_{\mathrm{OOD},k}) \,, \tag{4.1}$$

4.5 Control-relevant validity

Chapter 5

State of the art

5.1 Literature review strategy

5.2 Learned and hybrid models for vehicle dynamics control

5.3 Uncertainty-aware learned dynamics for control

5.4 OOD, distribution shift, and runtime monitoring

5.5 Validity-aware and confidence-aware control

5.6 Critical synthesis and research gap

Chapter 6

Conceptual framework and proposed architecture

6.1 The shift–validity–control chain

6.2 Hybrid prediction model

6.3 Validity estimator

6.4 Integration with MPC/NMPC

6.5 Optional extensions

6.5.1 Modulation of a CBF

6.5.2 Data acquisition and retraining

Chapter 7

Research and validation methodology (Priority 1)

7.1 Overall strategy

S1: Establish the control-oriented physical model and the baseline MPC/NMPC controller.

S2: Train the learned residual and quantify its improvement and limitations relative to the physical model within the nominal domain.

S3: Implement aleatoric and epistemic uncertainty estimation, calibration, and a computationally inexpensive OOD score.

S4: Design physically interpretable shifts in vehicle dynamics and characterize their effects on prediction, uncertainty, OOD, and validity.

S5: Construct and validate the signal C_k against control-relevant error.

S6: Integrate C_k into MPC/NMPC adaptation.

S7: Compare the baselines and the proposed approach under nominal and OOD conditions.

S8: Evaluate empirical robustness and, if progress permits, incorporate CBF and/or DAgger extensions.

7.2 Distribution shift scenarios

7.2.1 Unknown change in friction

7.2.2 Extension of the speed range

7.2.3 Sensor failure or degradation

7.2.4 Input OOD

7.3 Learned model and uncertainty

7.4 Construction and evaluation of the validity signal

7.5 Validity-aware integration with MPC/NMPC

7.6 Baselines and ablation study

B0 – Physics-only: Controller using a physical model without a learned residual.

B1 – Learned/no-awareness: Controller using a physical model and a learned residual, without validity estimation or OOD detection.

B2 – Validity/uncertainty-only: Controller using a learned model and a signal constructed from uncertainty/calibration, excluding an explicit OOD component.

B3 – OOD-only: Controller using a learned model and a response based solely on the OOD score, without probabilistic uncertainty estimation.

P – Proposed: Controller using a physical model, a learned residual, and the signal $C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Cal}, OOD)$, used to adapt the MPC/NMPC controller.

7.7 Evaluation metrics

7.7.1 Model metrics

7.7.2 Uncertainty, calibration, and OOD metrics

7.7.3 Closed-loop metrics

7.8 Empirical robustness criterion

$$\Delta J_{\text{prop}} < \Delta J_{\text{baseline}}, \tag{7.1}$$

7.9 Level of assurance and scope of the conclusions

7.10 Statistical analysis and reproducibility

Chapter 8

Preliminary results

8.1 Feasibility of learning vehicle dynamics models

8.2 Uncertainty and OOD behavior under distribution shifts

8.3 Preliminary integration with control

8.4 Discussion of preliminary results

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Expected contributions

9.1 Control-relevant validity estimation

9.2 Validity-aware vehicle control

9.3 Runtime validity characterization

9.4 Experimental taxonomy of shifts in vehicle dynamics

9.5 Potential extensions

Chapter 10

Work plan (Priority 1)

10.1 Remaining activities

10.2 Schedule

10.3 Decision milestones

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Risks, limitations, and mitigation strategies

11.1 Risk of redundancy between uncertainty and OOD

11.2 Risk of poor calibration or weak correlation with error

11.3 Risk that horizon adaptation does not improve closed-loop performance

11.4 Computational risk

11.5 Risk of scope expansion

11.6 Limitations of formal guarantees

Chapter 12

Conclusions of the qualifying examination

Bibliography

[1] Alberto Bertipaglia, Mohsen Alirezaei, Riender Happee, and Barys Shyrokau. Model predictive contouring control for vehicle obstacle avoidance at the limit of handling. In Wei Huang and Mehdi Ahmadian, editors, *Advances in Dynamics of Vehicles on Roads and Tracks III*, volume 2 of *Lecture Notes in Mechanical Engineering*, pages 132–142. Springer, 2024.

[2] Bosch. 30 years of ESP: technology developed by Bosch evolves and continues to shape the future of mobility. Bosch Media Service Brasil, 2025. Press release in Portuguese; title translated into English. Accessed October 12, 2026.

[3] J. Hu, H. Liu, S. Yi, and C. Huang. Lane-changing control strategy for distributed drive vehicles considering yaw stability. *IET Intelligent Transport Systems*, 19:e12605, 2025.

[4] Miren Isasa, Ángela Alonso-Solórzano, Itziar Gurrutxaga, and Heriberto Pérez-Acebo. A skid resistance predicting model for single carriageways. *Lubricants*, 13(8), 2025. Cited by: 4; All Open Access, Gold Open Access, Green Open Access.

[5] J. Rajendra Prasad, Shalini Kanuganti, Pooja N. Bhanegaonkar, Ashoke Kumar Sarkar, and Shriniwas Arkatkar. Development of relationship between roughness (iri) and visible surface distresses: A study on pmgsy roads. *Procedia - Social and Behavioral Sciences*, 104:322–331, 2013. 2nd Conference of Transportation Research Group of India (2nd CTRG).

[6] S. Salleh, R. Muhamad, and M.H. Abdillah. Performance of ralumac micro surfacing at blackspot eastbound, latar highway. volume 527, 2019. Cited by: 0; All Open Access, Gold Open Access.

[7] Viktor Skrickij, Paulius Kojis, Eldar Šabanovič, Barys Shyrokau, and Valentin Ivanov. Review of integrated chassis control techniques for automated ground vehicles. *Sensors*, 24(2):600, 2024.

[8] Zihang Weng, Zhen Leng, Yuchuan Du, Chenglong Liu, Xinqiang Zhang, Danning Li, and Yulin He. Leveraging tabular prior-data fitted networks for accurate pavement

friction coefficient prediction from 3d texture features. *Tribology International*, 218, 2026. Cited by: 4.

[9] D. J. Wilson. *An Analysis of the Seasonal and Short-Term Variation of Road Pavement Skid Resistance*. PhD thesis, The University of Auckland, Auckland, New Zealand, 2006.